



## **PROFESSIONAL PRACTICES**

### **ASSIGNMENT 4**

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## Question # (01)

### Reducing Errors in Computerized Hospital prescription System:-

In many modern hospitals, doctors use computerized systems to enter prescription orders for patients. These systems are very helpful because they reduce mistakes caused by unclear handwriting and automatically check for drug conflicts or allergies. However, technology can also create new types of errors if it is not used carefully. To reduce this kind of error, computerized hospital systems should include strong safety features.

#### Feature:

#### Automatic logout and session timeout:

One important feature



that should be included is an automatic logout or session timeout system. This feature would automatically log a doctor out of the system after a short period of inactivity, such as one or two minutes. This feature reduces the risk of another doctor using the same terminal under the wrong account.

## **Feature 2:**

### **Patient Identity Confirmation before order submission:**

Another useful feature is patient identity confirmation before order submission. Before a doctor submits a drug order, the system should clearly display patient's name, photograph, age and patient ID number.



## Conclusion:-

Although computerized prescription system greatly improve health care efficiency and reduce many traditional errors. They must be designed carefully, to avoid new risks. Feature such as automatic logout and patient identity confirmation can significantly reduces the chances of assigning medicines to the wrong patient.



Qo:02:-

## Cyber Laws and Ethical Handling in Pakistan-

### Cyber Crime Laws of Pakistan:

Cyber crime in Pakistan are primarily governed by the "

"Prevention of Electronic Crimes Act (PECA) 2016"

This Law addresses offenses such as online harassment, cyberstalking, identity theft, unauthorized access to data, electronic fraud, hate speech, and online threats. It defines punishments, investigations procedures and protects individuals from misuse of electronic systems.

### Initiatives Taken by Pakistan:

Pakistan has established the "FIA cyber crime wing" which investigates and prosecutes cyber crime cases nation wide. Online complaint portals, digital forensic units and public awareness campaigns have



been introduced to improve reporting of electronic crimes.

### Ethical and Responsible Response To an online Threat :-

if the threatening message is received from an unknown number the correct response is "not to reply or escalate the situation" the message should be preserved as evidence (URLs) and reported through the FIA cyber crime wing's official channels.

### Role of Temperament and Ethical Skills :-

Controlling one's temperament prevent impulsive reaction that could worsen the situation or create legal complications. this approach support conflict avoidance personal protection, and effective law enforcement involvement.

### Conclusion :-

Pakistan's cybercrime framework emphasizes legal reporting ethical behavior and digital conduct.

By remaining individuals protect themselves while contributing to a safe online environment.



## PROFESSIONAL PRACTICES - ASSIGNMENT # 4

### -: QUESTION No. 3 :-

#### -: Amusement Park Ride Control System :-

The design of a computer-controlled amusement park ride is a safety-critical application. Failures in such systems can lead to injury or loss of life, rather than only financial loss or inconvenience. Therefore, the development process must place safety as the highest priority, following established good software development procedures to minimize the likelihood of errors.

#### 1. Design and Development Phase:

To ensure system safety & robustness, the following measures must be applied during design and coding:

- Requirement Analysis & Constraints:

Clearly define operational limits. The software must enforce hard-coded safety constraints for MAX weight and balance. If readings exceed limits, the ride must not start.

- Redundancy and Diversity:

The system must avoid single points of failure by not relying on a single sensor or processor.



- Sensor Redundancy:

Multiple sensors per seat should verify occupancy and weight. Conflicting readings must trigger a safety error state.

- Voting Systems:

Independent computers should calculate balance and speed. The ride proceeds only if all systems agree.

- Fail-Safe Defaults:

The default system state must be safe. For example, brakes should require power to release so that power loss automatically engages them.

- Design for Failure:

The software must handle sensor disconnection or invalid data by preventing ride activation rather than failing unpredictably.

## 2. Technical Issue & Implementation:

- Input Validation: All sensor inputs must be validated to distinguish between passengers, objects, and faulty readings.
- Real-Time Response: A real-time Operating System (RTOS) is required to ensure immediate execution of critical control actions.
- Cybersecurity: The system should be isolated from external networks



(air-gapped) to prevent unauthorized access or tampering.

### 3. Testing and Verification:

- **Simulation Testing:** Simulations should cover edge cases such as uneven passenger distribution movement during operation, and sensor failures.
- **Independent Verification & Validation (IV&V):** An independent team must test the system to identify design or logic flaws.
- **Stress Testing:** The system should be tested beyond normal conditions, including maximum loads and extreme temperatures.

### 4. Operating Instructions & Human Factors:

Human involvement

remains essential during pre-start checks:

- **Manual Start Authorization:** A human operator must confirm passenger safety and press an enable button after automated checks.
- **Emergency Kill Switch:** A physical emergency stop must immediately cut power and engage breaks, independent of software.
- **Clear User Interfaces:** Operator displays should show specific error messages to prevent confusion and unsafe overrides.

### Conclusion:-

While software errors can't be entirely eliminated, disciplined design ensures failures result in a safe shutdown rather than catastrophic outcomes, fulfilling the ethical responsibility to protect human life.